Lab -ASP.NET - Downloading a Blob
To Begin with the Lab

Summary

In this lab, you learn how to download a blob (file) from an Azure Blob Storage Container to your local computer using a .NET application. The application connects to Azure Blob Storage, locates the required blob using a **BlobClient**, and downloads it to a specified local folder. This functionality is commonly used for retrieving documents, images, backups, and application files stored in Azure Storage.

Understanding

Azure Blob Storage stores files as **Blobs** inside a **Blob Container**. To download a file, the application first connects to the Azure Storage Account and accesses the required Blob Container. A **BlobClient** is then created for the blob that needs to be downloaded.
The DownloadToAsync() method copies the blob from Azure Storage to a specified location on the local computer. If the destination file already exists, it is replaced with the downloaded version. Downloading blobs allows applications to retrieve files whenever users request them.

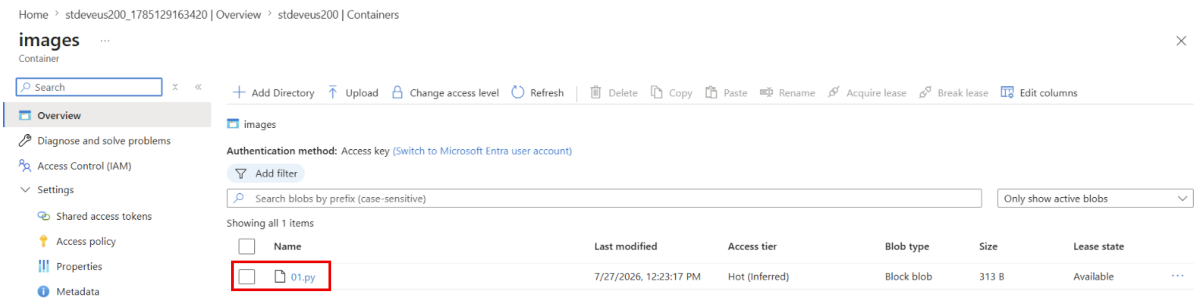
Example

Suppose your Blob Container contains the following file:
Storage Account:
mystorageaccount
Container:
images
Blob:
01.py
Download Location:
C:\Downloads\01.py
Application Flow:
Azure Storage Account
│
▼
Blob Container (images)
│
▼
01.py (Blob)
│
▼
BlobClient
│
▼
C:\Downloads\01.py
After the download completes, the file **01.py** is saved in the **C:\Downloads** folder.

Lab Steps

Step 1: Open the .NET Project

- Open your .NET project in Visual Studio or Visual Studio Code.
- Ensure the project is already connected to Azure Blob Storage.
- Verify that the blob you want to download already exists in the container.



Step 2: Create the Download Method

Create a method to download a blob from Azure Storage.

```
static async Task DownloadBlob(BlobContainerClient containerClient)
{
    string blobName = "01.py";
    string downloadPath = @"C:\Downloads\01.py";

    var blobClient = containerClient.GetBlobClient(blobName);

    await blobClient.DownloadToAsync(downloadPath);

    Console.WriteLine("Blob downloaded successfully");
}
```

- Specify the blob name.
- Specify the local Blowdownloads path.
- Create a BlobClient.
- Download the blob to the local computer.
- Display a success message.

powershell blobapp | Program.cs

blobapp > Program.cs

```csharp
static async Task UploadBlob(BlobContainerClient containerClient)
}

//await UploadBlob(containerClient);

static async Task ListBlobs(BlobContainerClient containerClient)
{
    await foreach(BlobItem blobItem in containerClient.GetBlobsAsync())
    {
        Console.WriteLine(blobItem.Name);
    }
}

//await ListBlobs(containerClient);

static async Task DownloadBlob(BlobContainerClient containerClient)
{
    string blobName = "01.py";

    string folder = @"C:\BlobDownloads";
    Directory.CreateDirectory(folder);   // Creates the folder if it doesn't exist

    string downloadPath = Path.Combine(folder, blobName);

    var blobClient = containerClient.GetBlobClient(blobName);

    await blobClient.DownloadToAsync(downloadPath);

    Console.WriteLine("Blob downloaded successfully");
}
```

Step 3: Call the Download Method

Call the method after creating the BlobContainerClient.

```
await DownloadBlob(containerClient);
```

- Starts the download process.
- Saves the blob to the specified location.

```csharp
static async Task DownloadBlob(BlobContainerClient containerClient)
{
    string blobName = "01.py";

    string folder = @"C:\BlobDownloads";
    Directory.CreateDirectory(folder);   // Creates the folder if it doesn't exist

    string downloadPath = Path.Combine(folder, blobName);

    var blobClient = containerClient.GetBlobClient(blobName);

    await blobClient.DownloadToAsync(downloadPath);

    Console.WriteLine("Blob downloaded successfully");
}

await DownloadBlob(containerClient);
```

Step 4: Build the Project

Run the following command:

```
dotnet build
```

- Compiles the application.
- Verifies that there are no build errors.

Expected Output:

Build succeeded.

Step 5: Run the Application

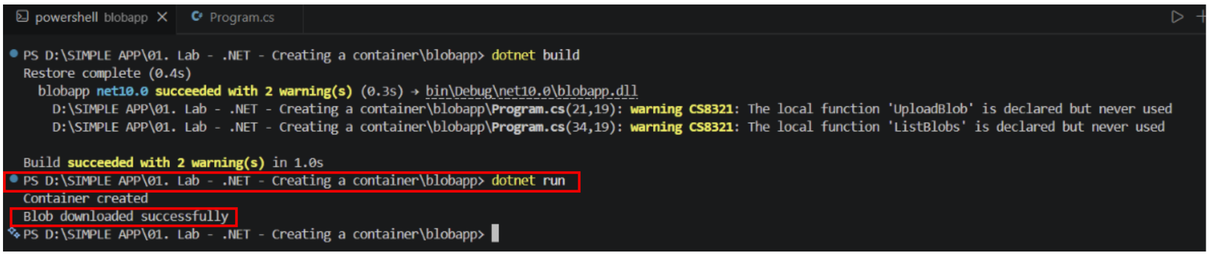
Run the following command:

```
dotnet run
```

- Starts the .NET application.
- Downloads the blob from Azure Storage.

Expected Output:

Blob downloaded successfully



Step 6: Verify the Downloaded File

- Open **File Explorer**.
- Navigate to the download folder.

Example:

C:\Downloads

- Verify that **01.py** has been downloaded successfully.
- Open the file to confirm its contents.